

Linear systems and elimination

Perturbed MILU conditioning for Neumann problems in two and three dimensions

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Controlling the perturbed factorization uniformly for singular Neumann systems in two and three dimensions is challenging. Community impact comes from a concrete preconditioner for widely used PDE linear solves.

Statement

Fix $d \in \{2, 3\}$ and a bounded connected smooth domain $\Omega \subset \mathbb{R}^d$. On $h\mathbb{Z}^d$, retain points whose cubic control cells $C_p = p + [-h/2, h/2]^d$ intersect Ω , ordered lexicographically as $1, \dots, N_h$. For face neighbors set

$$w_{pq} = \frac{\text{vol}_{d-1}(C_p \cap C_q \cap \Omega)}{h^{d-1}},$$

and set other weights to zero. Let $a_p = \sum_q w_{pq}$, $A_{pp} = a_p$, $A_{pq} = -w_{pq}$ for $p \neq q$, and let L be the strictly lower triangular part of A . This is the Purvis–Burkhalter Neumann discretization, with length fractions in two dimensions and face-area fractions in three dimensions.

For $\varepsilon > 0$, define $E = \text{diag}(e_1, \dots, e_{N_h})$ recursively by

$$e_1 = a_1, \quad e_p = (1 + \varepsilon)a_p - \sum_{\substack{q < p \\ w_{pq} > 0}} \frac{w_{pq}}{e_q} \sum_{s > q} w_{qs} \quad (p > 1),$$

and $M = (L + E)E^{-1}(E + L^T)$. This expresses the source’s PMILU recurrence by neighbor sums, including its explicitly unperturbed first pivot. Its two-dimensional display is Definition 5; §4 states the conjecture in both dimensions.

When the pivots are positive, let $\kappa_+(M^{-1}A)$ be the ratio of the largest to smallest positive eigenvalue of $M^{-1/2}AM^{-1/2}$; the zero eigenvalue from the Neumann nullspace is excluded.

Conjecture. There exist $c_\Omega, K_\Omega, h_\Omega > 0$, independent of h , such that choosing $\varepsilon = c_\Omega h^2$ gives positive pivots and

$$\kappa_+(M^{-1}A) \leq K_\Omega h^{-1} \quad (0 < h < h_\Omega).$$

The existence quantifier spells out the source’s choice of “some moderate constant”; it does not impose an invented interval for that constant. The authors suggest the practical choice $c_\Omega = 1$.

Numerical significance

The question concerns a concrete diagonal perturbation of incomplete factorization for singular Neumann systems. It is separate from relaxed ILU: the diagonal recurrence differs, and the proposed PMILU bound includes three dimensions.

References and status check

- B. Lee and C. Min, *Optimal preconditioners on solving the Poisson equation with Neumann boundary conditions*, J. Comput. Phys. **433** (2021), 110189, DOI. Author text: §2.1, equation (3); §4, Definition 5 and conjecture, pp. 5–6; §6.2 for the three-dimensional experiments.
- G. Hwang et al., *Localized Estimation of Condition Numbers for MILU Preconditioners on a Graph*, SIAM J. Numer. Anal. **64** (2026), 1443–1477, DOI; available preprint, §2.1 and §5, p. 21.

On 2026-09-10, checked the original full author text, the later preprint’s positive-definite assumptions and explicit Neumann future-work statement, its 2026 publication record, and targeted PMILU/Neumann/title searches including 2025–2026. No resolution was found. The 2026 journal full text was unavailable; this is a bounded check using its primary abstract and author preprint. The source’s descriptive phrase “MILU of $A + \varepsilon D$ ” differs at the first pivot from its displayed Definition 5; this entry follows the display.

Independent audit — 2026-09-10

Definition 5 and the §4 conjecture in the full Lee–Min author PDF support the displayed PMILU recurrence, including the first-pivot exception. The paper proves the rectangular RILU result, not a PMILU theorem; its smooth-domain PMILU evidence is numerical. The positive-definite assumptions and Neumann future-work boundary of the later preprint were checked, along with targeted subsequent searches. No exact proved subfamily warranting a partial-resolution label, or general resolution, was located.